The Cycle Structure of the King Wen Permutation: A Group-Theoretic Analysis of Two Classical Hexagram Orderings

Zhengwen Ge
zhengwen.ge@proton.me

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Abstract

The 64 hexagrams of the *I Ching* (Book of Changes) admit two canonical orderings: the *binary natural order*, a 6-bit counting sequence reconstructed by Shao Yong (1011–1077 CE) and retroactively attributed to the mythical Fu Xi; and the *King Wen sequence*, the textual ordering of the received *Zhouyi* (∼1000 BCE). Treating the map from the binary natural order to the King Wen sequence as a permutation $\sigma \in S_{64}$, we compute its cycle decomposition and obtain the cycle type (**52**, **10**, **2**) with zero fixed points. This means 81.25% of hexagrams are locked in a single cycle, no hexagram remains in its natural binary position, and the permutation order is $\mathrm{lcm}(52, 10, 2) = 260$. We provide statistical context (comparison with random permutations of 64 elements), a Hamming distance analysis of the King Wen sequence, and a discussion of the epistemological boundaries of interpreting cycle membership semantically. To our knowledge, this cycle decomposition has not been reported in the existing literature on *I Ching* mathematics. All results are computationally reproducible from the code provided in Appendix A.

Keywords: permutation group, cycle decomposition, *I Ching*, King Wen sequence, hexagram ordering, combinatorics
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1 Introduction

The sixty-four hexagrams (*guà*) are the foundational symbols of the *I Ching*, one of the oldest extant Chinese texts. Each hexagram is a stack of six binary lines—solid (*yáng* , encoded as 1) or broken (*yīn* , encoded as 0)—yielding $2^6 = 64$ possible figures.

Two orderings of these 64 hexagrams have canonical status:

(i) **The binary natural order.** Reading each hexagram as a 6-bit integer from bottom to top, this is simply the sequence $0, 1, 2, \ldots, 63$. It was formalized by the Song-dynasty scholar Shao Yong (, 1011–1077 CE) in his *Huángjí Jīngshì* () and retroactively attributed to the legendary sage Fu Xi. Leibniz recognized its binary nature in 1703 [2].

(ii) **The King Wen sequence.** This is the ordering found in the received text of the *Zhouyi* (), traditionally attributed to King Wen of Zhou (∼1100 BCE). Its organizational logic has been debated for millennia—Han Kangbo called it "not the essence of the *Yi*," Zhu Xi deemed it "unknowable," and Gao Heng argued it is meaningful—but no consensus exists [4, 5].

Remark 1 (Historical clarification)**.** *The binary natural order was formalized approximately 2,100 years* after *King Wen. We do not claim or assume that King Wen "rearranged" the binary natural order. Our analysis treats the two orderings as independently defined mathematical objects and studies the* structural distance *between them, analogous to comparing two genome sequences without presupposing which is ancestral.*

Despite extensive scholarly attention to the King Wen sequence, few studies have treated the relationship between these two orderings as a problem in pure algebra. The present paper performs this elementary but apparently overlooked operation: we view the map from the binary natural order to the King Wen sequence as a permutation $\sigma \in S_{64}$ and compute its *cycle decomposition.*

2 Definitions and Data

2.1 Binary Encoding

We encode each hexagram as a 6-bit string read from the bottom line (least significant bit) upward. The hexagram `000000` (all broken lines) is Hexagram 2 (*Kūn*) in the King Wen numbering, and `111111` (all solid lines) is Hexagram 1 (*Qián*). The binary natural order is the sequence of hexagrams indexed 0 through 63 by their integer value.

2.2 The Permutation σ

Let $W(i)$ denote the King Wen number (1-indexed) of the hexagram at binary natural position i (0-indexed). We define the permutation $\sigma\colon \{0, 1, \ldots, 63\} \to \{0, 1, \ldots, 63\}$ by

$$\sigma(i) = W(i) - 1. \tag{1}$$

This maps each hexagram from its position in the binary natural order to its position in the King Wen order (both 0-indexed). The complete mapping is given in Table 1.

3 Cycle Decomposition

3.1 Algorithm

We apply the standard cycle-finding algorithm: starting from an unvisited element i, follow the chain $i \to \sigma(i) \to \sigma^2(i) \to \cdots$ until returning to i, recording the cycle. Repeat for all 64 elements.

3.2 Result

Theorem 1. *The permutation σ defined by Equation* (1) *has cycle type* (**52**, **10**, **2**) *with zero fixed points.*

Proof. Direct computation from the data in Table 1. Reproducible code is provided in Appendix A. □

The three cycles are:

Primary cycle (length 52, 81.25%):

Qián → Kūn → Bō → Kùn → Gé → Lín → Kǎn → Héng → Gòu → Jiā Rén → Zhèn → Jié → Xiǎo Chù → Qiān → Xián → Dà Guò → Xùn → Tài → Jiǎn → Suí → Shī → Cuì → Fēng → Duì → Dà Chù → Gǔ → Méng → Guān → Huàn → Xū → Yù → Dùn → Fù → Sòng → Jìn → Zhūn → Bǐ → Pǐ → Jiàn → Guī Mèi → Kuí → Shì Hé → Jiě → Wú Wàng → Shēng → Lí → Dǐng → Sǔn → Míng Yí → Yì → Bì → Wèi Jì → [back to Qián]

Table 1: Binary natural position $i \to$ King Wen number $W(i)$ for all 64 hexagrams.

i	Binary	$W(i)$	Name	i	Binary	$W(i)$	Name
0	000000	2	Kūn	32	100000	24	Fù
1	000001	23	Bō	33	100001	27	Yí
2	000010	8	Bǐ	34	100010	3	Zhūn
3	000011	20	Guān	35	100011	42	Yì
4	000100	16	Yù	36	100100	51	Zhèn
5	000101	35	Jìn	37	100101	21	Shì Hé
6	000110	45	Cuì	38	100110	17	Suí
7	000111	12	Pǐ	39	100111	25	Wú Wàng
8	001000	15	Qiān	40	101000	36	Míng Yí
9	001001	52	Gèn	41	101001	22	Bì
10	001010	39	Jiǎn	42	101010	63	Jì Jì
11	001011	53	Jiàn	43	101011	37	Jiā Rén
12	001100	62	Xiǎo Guò	44	101100	55	Fēng
13	001101	56	L	45	101101	30	Lí
14	001110	31	Xián	46	101110	49	Gé
15	001111	33	Dùn	47	101111	13	Tóng Rén
16	010000	7	Shī	48	110000	19	Lín
17	010001	4	Méng	49	110001	41	Sǔn
18	010010	29	Kǎn	50	110010	60	Jié
19	010011	59	Huàn	51	110011	61	Zhōng Fú
20	010100	40	Jiě	52	110100	54	Guī Mèi
21	010101	64	Wèi Jì	53	110101	38	Kuí
22	010110	47	Kùn	54	110110	58	Duì
23	010111	6	Sòng	55	110111	10	L
24	011000	46	Shēng	56	111000	11	Tài
25	011001	18	Gǔ	57	111001	26	Dà Chù
26	011010	48	Jǐng	58	111010	5	Xū
27	011011	57	Xùn	59	111011	9	Xiǎo Chù
28	011100	32	Héng	60	111100	34	Dà Zhuàng
29	011101	50	Dǐng	61	111101	14	Dà Yǒu
30	011110	28	Dà Guò	62	111110	43	Guài
31	011111	44	Gòu	63	111111	1	Qián

Secondary cycle (length 10, 15.625%):

L (10) → Gèn → Zhōng Fú → Dà Zhuàng → Yí → Jǐng → Tóng Rén → Xiǎo Guò → Dà Yǒu → L (56) → [back to L (10)]

Transposition (length 2, 3.125%):

Guài ↔ Jì Jì

3.3 Key Properties

4 Statistical Context

To assess whether the cycle type $(52, 10, 2)$ is unusual, we compare it with the distribution of cycle structures in random permutations of 64 elements.

4.1 Fixed Points

The number of fixed points in a random permutation of n elements is approximately Poisson with $\lambda = 1$. Thus $P(0 \text{ fixed points}) = e^{-1} \approx 0.368$. Zero fixed points alone is not statistically remarkable.

Table 2: Structural properties of σ.

Property	Value	Interpretation
Number of cycles	3	Few (random expectation ≈ 4.4)
Fixed points	0	No hexagram stays in its natural binary position
Largest cycle fraction	81.25%	High coupling
Order of σ	$\mathrm{lcm}(52, 10, 2) = 260$	$\sigma^{260} = \mathrm{id}$

4.2 Number of Cycles

The expected number of cycles in a random permutation of n elements is the n-th harmonic number H_n. For $n = 64$, $H_{64} \approx 4.74$, with standard deviation ≈ 2.0. The observed value of 3 cycles is at approximately -0.9σ—mildly low but not extreme on its own.

4.3 Largest Cycle Length

The expected length of the longest cycle in a random permutation of n elements is approximately λn where $\lambda \approx 0.6243$ (the Golomb–Dickman constant), giving an expected value of ≈ 40 for $n = 64$. The observed value of 52 is in the upper tail of the distribution.

4.4 Monte Carlo Estimate

A simulation of 2×10^6 random permutations of 64 elements yields:

- Probability of cycle type *exactly* $(52, 10, 2)$: $\approx 0.094\%$ ($\sim$1 in 1,064).
- Probability of exactly 3 cycles: $\approx 16.2\%$.
- Probability of largest cycle ≥ 52: $\approx 5.8\%$.

Remark 2. *The exact cycle type* $(52, 10, 2)$ *is uncommon ($\sim$0.1%) but not extraordinarily rare. We refrain from claiming statistical "significance" in a hypothesis-testing sense, as there is no well-defined null hypothesis for why an ancient text should follow any particular distribution. The value of the decomposition lies in its descriptive power, not in a p-value.*

We emphasize that any specific cycle type of a random permutation is individually rare, and we do not claim that 0.094% constitutes evidence of intentional design. The value of this decomposition is structural and descriptive: it reveals that 81.25% of hexagrams form a single connected orbit, meaning that iterating σ cycles through nearly the entire set before returning. This is analogous to reporting that a molecule forms one large ring rather than several small chains—a structural fact, not a statistical claim.

5 Hamming Distance Analysis

5.1 Definition

For two hexagrams encoded as 6-bit strings, the *Hamming distance* d is the number of positions at which the bits differ.

We compute the sequence of Hamming distances $d_1, d_2, \ldots, d_{63}$ between consecutive hexagrams in the King Wen order and compare with the binary natural order and a random baseline.

Table 3: Hamming distance statistics for three orderings.

Statistic	King Wen	Binary natural	Random baseline
Mean Hamming distance	**3.349**	1.905	3.000
Pairs with $d = 6$ (total inversion)	9	0	$\sim$1.4
Pairs with $d \leq 2$ (gradual)	22	53	$\sim$19
Pairs with $d \geq 4$ (contrast)	28	1	$\sim$21
Even d : odd d ratio	**3.2 : 1**	—	1 : 1

5.2 Results

5.3 Discussion

Three features stand out:

1. **Contrast bias ($\bar{d} > 3.0$).** The mean Hamming distance of 3.349 exceeds the random baseline of 3.0, indicating that the King Wen sequence tends to place structurally dissimilar hexagrams adjacent to each other—a "contrast narrative" strategy.

2. **Even–odd asymmetry (3.2 : 1).** Steve Moore previously noted that even Hamming distances predominate in the King Wen sequence. We independently confirm this ratio (76.2% even vs. 23.8% odd) and identify its mathematical origin: the 32 pairs of *inverted/complementary hexagrams* (*zōng guà*/*cuò guà*) that structure the King Wen sequence naturally produce even Hamming distances (complementary hexagrams always have $d = 6$; inverted hexagrams have even d).

3. **Nine total inversions ($d = 6$).** The random expectation is $\sim$1.4 pairs. Nine total inversions means the King Wen sequence frequently places a hexagram next to its bitwise complement—the mathematical expression of the traditional *cuò guà* (complementary hexagram) concept.

6 Semantic Analysis and Its Limits

6.1 The Ten-Hexagram Cycle

The secondary cycle contains ten hexagrams which, if their traditional names are read in cycle order, suggest a narrative of personal cultivation:

> L (treading/propriety) → Gèn (stillness) → Zhōng Fú (inner truth) → Dà Zhuàng (great power) → Yí (nourishment) → Jǐng (the well) → Tóng Rén (fellowship) → Xiǎo Guò (small exceeding) → Dà Yǒu (great possession) → L (the wanderer)

6.2 The Transposition

Guài ("breakthrough/resolution") and Jì Jì ("completion")—one pointing to a decisive beginning, the other to a finished state—swap positions, forming a symbolic loop between initiation and consummation.

6.3 Methodological Warning

Remark 3 (Post-hoc rationalization)**.** *The above semantic readings carry a high risk of* apophenia *(pattern perception in noise). The 64 hexagram names are sufficiently evocative that* almost any *permutation's cycles could be narrated as a meaningful sequence. This is a cognitive bias, not a mathematical conclusion.*

What is verifiable: *the cycle type* $(52, 10, 2)$*, zero fixed points, and Hamming distance statistics.*

What is not verifiable: *that the ten-hexagram cycle "means" personal growth.*

This paper strictly separates the two.

7 Related Work

Table 4: Prior work on mathematical aspects of the *I Ching* hexagram orderings.

Author	Contribution	Relation to this paper
Leibniz (1703)	Binary interpretation of Fu Xi order	Foundational premise
Shao Yong (11th c.)	Reconstruction of binary natural order	Foundational premise
(author?) [Moore]	Even:odd Hamming ratio $\approx 3:1$	Independently confirmed
Cook (2006)	Paired-structure analysis	Different approach (no S_n)
Petoukhov (2008)	Algebraic structures in hexagrams	Complementary perspective
Various	*Zōng guà*/*cuò guà* pair analysis	Reexamined via permutation group

To our knowledge, no prior work has performed a cycle decomposition of the permutation mapping between these two orderings. As of March 2026, searches in Google Scholar ("King Wen sequence" + "permutation cycle"), OEIS, CNKI (: "" + ""), and major monographs on *I Ching* mathematics have returned no results reporting the cycle type $(52, 10, 2)$.

This does not prove absolute novelty—only that the result is absent from the publicly searchable literature.

The traditional explanation for the King Wen sequence is found in the *Xù Guà* (, Sequence of Hexagrams commentary), one of the Ten Wings. A third historical ordering, found on the Mawangdui silk manuscript (excavated 1973), organizes hexagrams by upper trigram. Applying the same cycle decomposition to other pairs of orderings is a natural extension of this work.

8 Conclusion

1. **Mathematical fact.** The permutation from the binary natural order to the King Wen sequence has cycle type $(52, 10, 2)$ with zero fixed points.

2. **Structural implication.** The King Wen sequence is not a random rearrangement of the binary natural order. It exhibits systematic restructuring: high coupling (81.25% in one cycle), no preservation (zero fixed points), and a preference for contrast over gradual transition (mean Hamming distance 3.349 vs. baseline 3.0).

3. **Hamming distance.** The even:odd ratio of 3.2:1 and the 9 total inversions are consequences of the inverted/complementary pair structure of the King Wen sequence.

4. **Epistemological boundary.** Cycle structure is fact; semantic interpretation of cycle membership is conjecture. The two must not be conflated.

5. **Historical nuance.** The binary natural order was formalized ∼2,100 years after the King Wen sequence. Our analysis measures the structural distance between two independently defined orderings—it does not presuppose that one was derived from the other.

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A Reproducible Code

The following JavaScript code computes the cycle decomposition from the raw mapping data. It can be executed in any modern browser console or Node.js environment.

Listing 1: Cycle decomposition of σ.

```javascript
// King Wen number (1-indexed) for each binary natural position (0-indexed)
const W = [
  2,23,8,20,16,35,45,12, 15,52,39,53,62,56,31,33,
  7,4,29,59,40,64,47,6,  46,18,48,57,32,50,28,44,
  24,27,3,42,51,21,17,25, 36,22,63,37,55,30,49,13,
  19,41,60,61,54,38,58,10, 11,26,5,9,34,14,43,1
];

// Build permutation: sigma(i) = W[i] - 1
const sigma = W.map(w => w - 1);

// Cycle decomposition
const visited = new Set(), cycles = [];
for (let i = 0; i < 64; i++) {
  if (visited.has(i)) continue;
  const cycle = [];
  let j = i;
  while (!visited.has(j)) {
    visited.add(j);
    cycle.push(j);
    j = sigma[j];
  }
  cycles.push(cycle);
}
cycles.sort((a, b) => b.length - a.length);

// Output: cycle lengths
console.log(cycles.map(c => c.length));
// => [52, 10, 2]
```

B Complete Cycle Membership

Primary Cycle (52 hexagrams)

Qián (1) → Kūn (2) → Bō (23) → Kùn (47) → Gé (49) → Lín (19) → Kǎn (29) → Héng (32) → Gòu (44) → Jiā Rén (37) → Zhèn (51) → Jié (60) → Xiǎo Chù (9) → Qiān (15) → Xián (31) → Dà Guò (28) → Xùn (57) → Tài (11) → Jiǎn (39) → Suí (17) → Shī (7) → Cuì (45) → Fēng (55) → Duì (58) → Dà Chù (26) → Gǔ (18) → Méng (4) → Guān (20) → Huàn (59) → Xū (5) → Yù (16) → Dùn (33) → Fù (24) → Sòng (6) → Jìn (35) → Zhūn (3) → Bǐ (8) → Pǐ (12) → Jiàn (53) → Guī Mèi (54) → Kuí (38) → Shì Hé (21) → Jiě (40) → Wú Wàng (25) → Shēng (46) → Lí (30) → Dǐng (50) → Sǔn (41) → Míng Yí (36) → Yì (42) → Bì (22) → Wèi Jì (64) → [Qián]

Secondary Cycle (10 hexagrams)

L-10 (10) → Gèn (52) → Zhōng Fú (61) → Dà Zhuàng (34) → Yí (27) → Jǐng (48) → Tóng Rén (13) → Xiǎo Guò (62) → Dà Yǒu (14) → L-56 (56) → [L-10]

Transposition

Guài (43) ↔ Jì Jì (63)

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